

# Grace Zhu

Seattle, WA, 98121, United States | 6478791809 | [zhuxingrace0115@gmail.com](mailto:zhuxingrace0115@gmail.com)

## Summary

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Senior Data Engineer with 8 years of experience driving cloud data and automation projects. Specializes in building scalable data pipelines and robust data services using Java, Python, SQL, and big data technologies in cloud environments. Adept at cross-functional collaboration, delivering quality features, and applying engineering best practices to deliver strategic technical impact.

## Core Skills

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- **Cloud & Data Platforms:** AWS (S3, EC2, Kinesis), Snowflake, Data Lakes, Real-time Streaming (Kafka)
- **Integration & APIs:** RESTful API, GraphQL, gRPC, Microservices, SSO (SAML, OAuth2)
- **Automation & CI/CD:** Apache Airflow, Jenkins, GitLab CI/CD
- **Data & Analytics:** SQL, Python, Time-Series Analysis, Anomaly Detection
- **Collaboration & Communication:** Cross-functional Leadership, Requirements Definition, Executive Reporting

## Work Experience

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### Royal Bank of Canada

Sep 2022 - Present

*Senior Software Developer - Data Engineering*

*Toronto*

Directed cloud data and automation projects with 8 years of expertise, emphasizing cross-functional collaboration, requirements definition, and end-to-end delivery in alignment with evolving data platform practices.

- Oversaw the migration of the enterprise HR data warehouse from Teradata to Snowflake, enhancing scalability, performance, and the underlying data platform architecture to support robust analytics and reporting capabilities.
- Managed the complete delivery of a Workday Data Pipeline in Java, integrating HR feeds into AWS S3 and Snowflake while tracking ETL workflow KPIs, which improved downstream analytics availability by 35%.
- Automated ServiceNow ticket creation using Python and the Snowflake Connector, reducing manual intervention in data mismatch resolution by 40% and reinforcing streamlined data processing practices.
- Coordinated a cross-functional rollout of an AI-driven HR inquiries automation program, collaborating with Data Science, Legal, and Support teams to reduce call volume by 45% and secure \$1.65M in annual savings, demonstrating leadership in data service innovation.
- Led the integration initiative for the Phenom Talent Experience Platform, partnering with Product, HR, and Vendor teams to automate 80% of candidate interactions, reduce time-to-fill by 30%, and enhance real-time data reporting, emphasizing strategic data insights.
- Mentored junior engineers and established governance rituals through multiple sprints, reinforcing engineering best practices for design reviews, coding standards, and risk management, all within the framework of a robust data engineering lifecycle.

### Royal Bank of Canada

Dec 2017 - Sep 2022

*Data Engineer*

*Toronto*

- Developed scalable backend services with Java Spring Boot to aggregate and process client data from multiple sources, laying the groundwork for robust data pipeline frameworks.
- Engineered and optimized REST APIs to integrate client financial data into internal systems, supporting high concurrency and real-time analysis while applying cloud computing and API development best practices.
- Applied Python for time series analysis on client performance data to build a backend system that efficiently processed insights, aligning with high standards for data service performance and reliability.
- Automated data workflows using Python and SQL, streamlining data collection, transformation, and storage processes to reduce latency and reinforce operational efficiency in a dynamic data environment.

## Education

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### University of Toronto

Sep 2013 - Jun 2017

*Bachelor of Science, Statistics*